

KUSHITHA LAKSHITHA ABAYAWICKRAMASINGHE

Embedded System | IoT Development |

Electronics Design | Robotics Enthusiast



kushithalakshitha.live



0714807853



in/kushitha-lakshitha/



Sri Lanka



github.com/Kushitha1521



kushithalakshitha@gmail.com



SUMMARY

Final-year Electronics & Computer Science undergraduate at the University of Kelaniya with strong skills in electronics, embedded systems, digital chip design using Altera FPGA boards, and laser communication systems. Experienced in IoT development with microcontroller-based systems and Arduino, and full-stack web development using React, Next.js, Spring Boot, and FastAPI. A cybersecurity enthusiast with interests in AI and cloud technologies, passionate about developing innovative real-world technology solutions.

EDUCATION

BSc(Hons) Electronics and Computer Science

GPA 3.6

FINAL YEAR UNDERGRADUATE

UNIVERSITY OF KELANIYA

COMPUTER HARDWARE CERTIFICATE COURSE

ICT CENTER - UNIVERSITY OF KELANIYA

MAKERSPACE PROGRAM (ELECTRONICS, ROBOTICS)

AMERICAN CENTER - U.S. EMBASSY IN SRI LANKA

G.C.E A/L EXAMINATION (2021)

TELIJJAWILA CENTRAL COLLEGE

Passed with results of A, B, C in the physical science subject (Com. math, ICT, Physics)

EXPERIENCE

PROJECT AREA COORDINATOR

2022 (6 MONTH)

- World Food Programme (WFP), Sarvodaya Shramadana Movement, Matara

PROJECTS

On Going

LASER-BASED FREE SPACE OPTICAL LINK FOR HIGH-SPEED DIGITAL DATA TRANSMISSION

09/2025-NOW

- Researching high-speed digital data transmission using laser-based FSO links, focusing on signal integrity, efficiency, and real-time monitoring.

Technologies: ESP32, Arduino Nano, FPGA Board, Photodiodes, Lasers, TIA, OpAmps, UART,

Completed

COMBAT ROBOT WITH WEAPON SYSTEM - UOK ROBOT BATTLES 2K25

09/2025-07/2025

- Designed and developed a remote-controlled combat robot ("Tombstone") with an integrated weapon system, focusing on motor control, wireless communication, and power management.

Technologies: ESP32, Arduino, Motor Drivers, 2.4GHz MC6C Transmitter & MC7RB Receiver, Li-Po Battery, DC Motors (300/550 RPM), Relay Module

TECHNICAL SKILLS

Electronics & IoT - Embedded Systems Development, Simulation & PCB Design, UART, SPI & I2C Communication, Firmware Development, Sensor Interfacing, Digital Logic Design, 3D Modeling, Troubleshooting & Debugging Hardware

Tools & Platforms - FPGA Boards, ESP32, STM32, Arduino Nano & Uno, PIC Microcontrollers, Proteus, Altium Designer, Altera Quartus Prime, Fusion 360

Languages - Embedded C/C++, Java, Verilog HDL, Python, C#, PHP, SQL, HTML, CSS, JavaScript,

AI & Data Science - TensorFlow, Hugging Face, NumPy, Pandas, Scikit-learn

Web & Cloud Technologies - Azure, Vercel, Docker, Git, GitHub Actions, Supabase, MySQL, MongoDB, FastAPI, React.js, Tailwind CSS, Spring Boot

Cybersecurity & Networking - Linux, Burp Suite, Wireshark, Postman, Nmap, Ethical Hacking, CTF

CERTIFICATION

- Computer Hardware
- Cyber Threat Intelligence - Hashx

LEADERSHIP & TEAMWORK

TEAM LEADER 24/25

TECHNICIAN 23/24

- Equipment Repair Unit (ERU), University of Kelaniya

DIGITAL MARKETING LEAD 23/24

DIGITAL MARKETING LEAD (JUNIOR) 22/23

- E-WASTE PROJECT, Electronic Development and Innovation Center (EDIC), University of Kelaniya

FINANCE COORDINATOR

22/23

- AWS Club, University of Kelaniya

LEAD THE WEB DEVELOPMENT TEAM

2024

- Humane Society Web Project, Humane Society, University of Kelaniya
- React.js, Tailwind CSS and Vite



WEB DEVELOPER

2024

- Medusa 1.0 website (TEAM WORK), Electronic and Computer Science Club, University of Kelaniya
- React.js, Tailwind CSS and Vite



ENERGY MONITORING & CONTROLLING SYSTEM

05/2025-09/2025 

- Developed an IoT monitor using ESP32 to track real-time home power consumption and ESP01 to control the system.
- Integrated an AI model to analyze usage history and predict future energy costs for users.

Technologies: Supabase, Hugging Face, Proteus, Arduino, Docker, ESP32, ESP01, ACS712, ZMPT101B

4-BIT ALU DESIGN AND IMPLEMENTATION

2026 

- Developed a 4-bit ALU using Verilog HDL on Altera MAX II CPLD, supporting arithmetic and logical operations with status flags. Followed complete digital design flow including RTL modeling, simulation, synthesis, and hardware validation.

Technologies: Verilog HDL, Altera MAX II CPLD, Quartus Prime Lite

PORTABLE 18650 LITHIUM-ION POWER BANK

10/2025-03/2026 

- Designed a portable power bank system with safe lithium battery charging and integrated protection circuitry, delivering a stable 5V output; circuit designed and simulated using Altium Designer.

Technologies: TP4056, BMS Protection, and XL6009 Boost Converter, Altium Designer, 18650 battery

ROBOTIC ARM 3D DESIGN & SIMULATION

04/2026-05/2026 

- Designed a 6DOF robotic arm 3D model using Fusion 360, focusing on joint movement, kinematics, and mechanical structure for future embedded and automation integration

Technologies: Fusion 360, CAD Design, Robotics, Mechanical Systems, Kinematics

EV CHARGING STATION RESERVATION SYSTEM

09/2025-NOW 

- Developing a full-stack EV charging platform with real-time slot booking, secure payments, and a user-friendly dashboard for managing schedules and charging history.

Technologies: Spring Boot, React, TypeScript, Tailwind CSS, Vite, MongoDB, Figma, Stripe, JWT

REAL-TIME CRYPTO SIGNAL PLATFORM

02/2025-06/2025 

- Built a live crypto dashboard with a signal engine using indicators like MACD and RSI to deliver real-time Buy/Sell insights from Binance, deployed on Microsoft Azure for reliable data streaming and performance.

Technologies: React, Tailwind CSS, Vite, Flask, Python, Binance API, Azure

WEB DEVELOPER

2024

- UOK ROBOT BATTLE website (TEAM WORK), Electronic and Computer Science Club, University of Kelaniya
- React.js, Tailwind CSS and Vite



WORDPRESS DEVELOPER

2024

- TECH HIVE PROJECT, Zero Plastic Movement, University of Kelaniya

DESIGNER

22/23

- BR AVENUE, Leadership Empowerment Development(LED), University of Kelaniya

DESIGNER

22/23

- Electronics' Student Society(ESS), University of Kelaniya

ACHIEVEMENTS

- 2nd Runner-up – UOK Robot Battles 2K25
- 1st Runner-up – JapuraExpo 2K25
- 1st Runner-up – ICAPS 2025 Undergraduate Research Symposium (UOK)
- Participated in the final round - Medusa 1.0 (2024)(CTF Competition), University of Kelaniya
- A team member of one of the top ten teams - PARAMOUNT 3.0 Mathematics Competition (2024), University of Kelaniya

INTERESTS

- Karate
- Photography
- Graphic Design & Video Editing
- Digital Marketing

LANGUAGE

- English
- Sinhala

REFERENCE

Dr. Jehan Senavirathne
Senior Lecturer, Faculty of Science,
University of Kelaniya, Sri Lanka
jehans@kln.ac.lk
+94718226117

Geeth Priyanka
Senior Software Engineer
Sysco LABS, Sri Lanka
070 257 6113
Kandy